Anytime-Valid Nonparametric Testing of Synergistic (Group) Causality, with an Application Establishing Its Absence in Daily Financial Markets

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Abstract

Does a *group* of causes act on an effect jointly—beyond the sum of its members' separate contributions—and can we decide this *as data arrive*, stopping the moment the evidence is conclusive? Every existing test of such synergy is fixed-sample: sufficient-cause interaction indices, joint-effect estimators, and information-theoretic decompositions (PID, SURD, PEID) all return a point value at a pre-committed sample size, with no valid way to monitor evidence continuously. We introduce ANTE, an *anytime-valid* nonparametric test of a super-additive synergy null built on testing-by-betting. A bounded, mean-centred interaction score is accumulated into a nonnegative capital process (an e-process); Ville's inequality yields time-uniform Type-I error control under optional stopping and continuous monitoring, requiring neither stationarity nor correct model specification. We prove validity, a growth (consistency) theorem giving power one and an explicit detection-time bound under the alternative, and extend the construction to $\boldsymbol{k}$-way groups and to an out-of-sample predictive (Granger) synergy for time series. Empirically, ANTE holds its level under continuous monitoring where a peeking fixed-sample test inflates tenfold, matches the best batch estimators in detection accuracy, and—uniquely—isolates genuine interaction from mere joint dependence. Deployed as a demonstrably powered instrument, ANTE *establishes* that super-additive group causality is *absent* from daily financial markets across thousands of asset triples, while it detects the turbulent energy cascade in physics. Synergistic group causality is real, but it is not a daily-markets phenomenon.

Keywords: anytime-valid inference, e-processes, testing by betting, synergy, group causality, Granger causality

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1 Introduction

Some effects require a coalition. Two drugs that are inert alone can be lethal together; two genes silent in isolation can jointly switch on a phenotype; two market shocks, each absorbed on its own, can compound into a crisis. The common structure is *synergy*: the joint action of a set of causes exceeds the sum of their individual actions. Detecting synergy—distinguishing "you need *both* A and B" from "A helps and B helps, separately"—is a recurring scientific question, and a surprisingly subtle one, because it is a statement about an *interaction*, not about any single cause.

The question has been formalised at least three times, in mutually unaware literatures. In epidemiology, Rothman's sufficient-component-cause model (Rothman 1976) defines synergy as two factors being component causes of the same causal "pie", measured by the additive-scale interaction contrast and its relative-excess-risk index (Hosmer and Lemeshow 1992; VanderWeele and Knol 2014), with a complete counterfactual axiomatisation for arbitrarily many binary causes due to VanderWeele and Richardson (2012) and, very recently, a multi-factor generalized synergy index (La Torre and D'Urso 2026). In statistics, a parallel programme estimates the effect of a *joint* intervention $\mathrm{do}(\{A, B, \dots\})$ by fusing experiments (Roquero Gimenez and Rothenhäusler 2023) or identifying the joint law of potential outcomes (Wu and Mao 2025). In information theory, Partial Information Decomposition (Williams and Beer 2010) and its dynamical descendants SURD (Martínez-Sánchez et al. 2024) and PEID (Yang et al. 2026) split the information that sources carry about a target into redundant, unique, and *synergistic* parts.

All three share one limitation that this paper is written to remove: they are **fixed-sample**. Each returns a point estimate—and, at best, a confidence interval—at a sample size chosen *before* looking at the data. None supports the mode of inference that modern data actually demand: watch the evidence accumulate, and stop as soon as it is decisive, without paying a multiplicity penalty for having looked. A practitioner monitoring a streaming system who applies a fixed-sample synergy test repeatedly, "peeking" after each new observation, silently destroys the guarantee the test was built to provide; we show below that the Type-I error of exactly this natural procedure inflates from a nominal 5% to over 50%.

The machinery to do continuous monitoring correctly now exists. Testing by betting (Shafer 2021), e-processes and safe anytime-valid inference (Ramdas et al. 2023; Grünwald et al. 2024), and confidence sequences (Howard et al. 2021) provide tests whose validity holds *simultaneously at all sample sizes* and under *any* data-dependent stopping rule. The idea is disarmingly simple: a sceptic bets against the null; if the null is true the sceptic's capital is a nonnegative martingale and cannot grow, so large capital is itself evidence against the null, quantified by Ville's inequality (Ville 1939). This programme has produced anytime-valid tests for means (Waudby-Smith and Ramdas 2024), off-policy and average treatment effects (Waudby-Smith et al. 2024), and other structured nulls—but, to our knowledge, *never for an interaction or synergy contrast.* The two bodies of work—synergy on one side, anytime-valid inference on the other—have not met.

Contributions.

We make them meet. This paper contributes:

1. **A test.** ANTE (*Anytime-valid Nonparametric Test of synErgy*): a betting-martingale test of a super-additive synergy null. Under a randomized or known design we build a bounded interaction score that is exactly mean-centred under H_0 (Lemma 1); accumulating it as capital yields a nonnegative martingale (Theorem 2) and, via Ville's inequality, time-uniform Type-I control under arbitrary optional stopping (Corollary 3). Validity needs no asymptotics, no variance assumption, and no stationarity.
2. **Theory of power.** We prove that under any fixed super-additive alternative the capital grows exponentially, so the test rejects with probability one (Theorem 4), and we give an explicit high-probability bound on the detection time (Proposition 5). This is the guarantee a fixed-sample estimator cannot offer and the property a Q1 audience rightly demands.
3. **Groups and time series.** We extend the score to k-way groups by contrasting a joint predictor carrying all higher-order interaction terms against a purely additive one (Proposition 6), and we instantiate a sequential test of *synergistic Granger causality*—super-additive out-of-sample predictive gain—for nonstationary time series, using forgetting recursive-least-squares predictors and a game-theoretic (prequential) null that dispenses with correct specification.
4. **A discrimination guarantee.** We show, and verify, that ANTE separates genuine irreducible interaction from two impostors that fool existing estimators: additive-but-both-needed dependence, and within-variable nonlinearity $Y = g(A)$ (Proposition 7).
5. **A scientific finding.** Turning the test on real data, we *establish*—as evidence of absence, not absence of evidence—that super-additive group causality is absent from daily financial markets. Across more than three thousand asset triples spanning returns, volatility and tail-stress, daily and hourly, equities, sector ETFs, macro assets and crypto, and 2-, 3- and 4-way groups, the financial evidence sits on a calibrated true-null distribution (median $\log_{10} E = -0.93$: a bet on synergy systematically loses), with zero family-wise detections. The very same test detects synthetic synergy at 74–100% power and recovers the turbulent energy cascade from fluid dynamics. Daily markets are common-factor plus pairwise; their irreducible higher-order structure is negligible—a quantified caution to in-sample synergy-in-finance claims (Scagliarini et al. 2020).

The paper is organised as follows. Section 2 positions ANTE against the three synergy literatures and the betting backbone. Section 3 fixes the model and defines the synergy null on two scales. Section 4 is the theoretical core: score, e-process, validity, power, groups, and the isolation guarantee. Section 5 gives the algorithms. Section 6 reports synthetic validation, a head-to-head benchmark against 2024–2026 methods, and the real-data study culminating in the evidence-of-absence finding. Sections 7 and 8 discuss and conclude; proofs are collected in the appendix.

2 Related work and positioning

We survey the three traditions that formalise group causality and the sequential machinery we bring to bear, then state precisely the gap ANTE fills.

2.1 Sufficient-cause and additive-scale interaction

Rothman's causal-pie model (Rothman 1976) is the oldest formalisation of "A and B together cause C". For two binary causes with cell means $\mu_{ab} = \mathbb{E}[C \mid \mathrm{do}(A{=}a, B{=}b)]$, synergy on the additive (risk-difference) scale is the interaction contrast $\theta = \mu_{11} - \mu_{10} - \mu_{01} + \mu_{00}$, whose sampling theory is the RERI literature (Hosmer and Lemeshow 1992; VanderWeele and Knol 2014). VanderWeele and Richardson (2012) give the definitive counterfactual theory of sufficient-cause interaction for arbitrarily many dichotomous exposures; La Torre and D'Urso (2026) extend Rothman's two-factor synergy index to many factors. These are exactly the batch nulls ANTE generalises—but every one is a point estimate and confidence interval at fixed n, and the authors themselves note instability of multi-way interaction at small n, precisely the regime where a sequential test that spends evidence adaptively helps most.

2.2 Joint effects and causal aggregation

A second tradition estimates the effect of a joint intervention rather than testing for interaction. Roquero Gimenez and Rothenhäusler (2023) fuse experiments that each manipulate a few variables to recover joint-intervention effects; Wu and Mao (2025) identify the joint distribution of potential outcomes from multiple experiments. These target the joint *effect*, again one-shot, with no time-uniform coverage—even though the single-treatment anytime-valid literature is mature (Waudby-Smith et al. 2024).

2.3 Information-theoretic synergy

PID (Williams and Beer 2010) decomposes the information two sources carry about a target into redundant, unique and synergistic atoms. SURD (Martínez-Sánchez et al. 2024) recasts this for causal, dynamical systems, robust to nonlinearity and colliders; PEID (Yang et al. 2026) gives an interventional analogue whose maximum-entropy-intervention identity makes source-side redundancy vanish, so synergy reads off as $EI(A, B \to C) - \sum_i EI(T^{(i)} \to C)$, and whose *hierarchical additivity* supplies the increment structure our subset-refinement search exploits. A cautionary result—the PID lattice becomes inconsistent for four or more variables (Lyu et al. 2026)—motivates a *test*-based rather than lattice-based treatment. Critically, all three are *estimators*: they return a synergy value with no null distribution, p-value, or stopping rule.

2.4 Testing by betting and anytime-valid inference

Our backbone is game-theoretic statistics (Ramdas et al. 2023): Ville's inequality (Ville 1939), the betting reframing of testing (Shafer 2021), safe testing (Grünwald et al. 2024), the hedged-capital / GRAPA betting strategies of Waudby-Smith and Ramdas (2024), and e-value multiplicity control (Vovk and Wang 2021; Wang and Ramdas 2022). The closest causal application is anytime-valid off-policy and ATE inference (Waudby-Smith et al. 2024); like all prior work it targets a *single* effect, not an interaction.

2.5 The gap, in one table

Table 1 The two literatures do not intersect. No prior method is both a synergy test and anytime-valid.

Literature	Handles synergy?	Anytime-valid / sequential?
Sufficient-cause / RERI / synergy index (§2.1)	yes	no (fixed n)
Joint effects / causal aggregation (§2.2)	yes (joint effect)	no (one-shot)
PID / SURD / PEID (§2.3)	yes (info scale)	no (point estimate)
Betting / e-processes / anytime-valid ATE (§2.4)	no (single effects)	yes
ANTE (this paper)	**yes**	**yes**

A focused prior-art search (terms: *anytime-valid interaction test*, *e-process causal interaction*, *sequential test synergy*, *confidence sequence interaction effect*) found no sequential, anytime-valid test whose null is "the joint causal effect of a set of causes equals the sum of its parts." ANTE is, to our knowledge, the first.

3 Problem setup and the synergy null

We develop the null on two scales—an interventional additive-interaction scale and an observational predictive scale—and show they reduce to the same qualitative statement: *joint equals sum of parts.*

3.1 Interventional synergy

Observations arrive as a stream $Z_1, Z_2, \ldots$ with $Z_t = (X_t, \mathbf{T}_t, C_t)$, where $C_t \in [0,1]$ is a bounded effect at a target, $\mathbf{T}_t = (T_t^{(1)}, \ldots, T_t^{(k)}) \in \{0,1\}^k$ are candidate binary causes, and X_t are optional covariates. Let $\mathcal{F}_t = \sigma(Z_1, \ldots, Z_t)$ be the natural filtration. We adopt the potential-outcomes model with $C_t(\mathbf{t})$ the outcome under $\mathrm{do}(\mathbf{T} = \mathbf{t})$ and cell means $\mu_{\mathbf{t}} = \mathbb{E}[C(\mathbf{t})]$.

Assumption 1 (Design) Per step, (i) *consistency* $C_t = C_t(\mathbf{T}_t)$; (ii) *positivity* $\pi(\mathbf{t} \mid X) \geq \pi_{\min} > 0$; (iii) *unconfoundedness* $\mathbf{T}_t \perp \{C_t(\mathbf{t})\} \mid X_t$. Under a randomized or known design $\pi(\mathbf{t} \mid X) = \pi(\mathbf{t})$ is known and (i)–(iii) hold by construction.

Definition 1 (Additive interaction contrast) For two causes A, B,

$$\theta \;=\; \mu_{11} - \mu_{10} - \mu_{01} + \mu_{00}.$$

$\theta = 0$ states that the joint effect of (A, B) equals the sum of the two individual effects (*no synergy*); $\theta > 0$ is super-additive synergy (a causal-pie / AND mechanism), $\theta < 0$ is antagonism. The one-sided synergy null is $H_0 : \theta \leq 0$ (two-sided $H_0 : \theta = 0$).

The information-theoretic synergy of §2.3 targets the same idea in bits: under an independent maximum-entropy intervention, source-side redundancy vanishes and $\mathrm{Syn}_{EI} = EI(A, B \to C) - \sum_i EI(T^{(i)} \to C)$ (Yang et al. 2026). Both scales reduce to *joint = sum of parts*; §4 develops the exactly-identified, bounded additive-scale test in full and remarks on the information-scale instantiation.

3.2 Observational (Granger) synergy for time series

For financial and other observational streams there is no intervention, so we move to an out-of-sample predictive definition in the spirit of Granger (Granger 1969). Let the (standardized) panel have target Y, candidate sources A, B, optional conditioning set Z (e.g. a market factor), and lag order p. For $S \subseteq \{A, B\}$ let $L(S)$ be the expected one-step-ahead loss of the best predictor of Y_t from the $\{Y, Z\}$-past augmented by the lagged histories of S, and let $\Delta(S) = L(\varnothing) - L(S)$ be its predictive gain.

Definition 2 (Predictive (Granger) synergy)

$$\mathrm{Syn}(A, B \to Y) \;=\; \Delta(\{A, B\}) - \Delta(\{A\}) - \Delta(\{B\}).$$

$\mathrm{Syn} > 0$ means the joint predictor beats the sum of the marginal predictors: the pair carries predictive structure neither member carries alone.

For Gaussian variables, Granger causality equals transfer entropy (Barnett et al. 2009), so Syn is an interaction of conditional transfer entropies—the same object SURD/PID call synergistic information, recast as an out-of-sample predictive contrast. The advantage of the predictive framing is that it supports a *prequential* null (Dawid 1984) requiring neither stationarity nor correct specification, as we now make precise.

4 The ANTE test

4.1 A per-sample score, centred under the null

Under Assumption 1 define the inverse-probability-weighted *interaction score*

$$\psi_t \;=\; c(A_t, B_t)\, \frac{C_t}{\pi(A_t, B_t)}, \qquad c(a, b) = (-1)^{(1-a)+(1-b)}, \tag{1}$$

so that $c(a,b) = +1$ on the concordant cells $(a,b) \in \{(1,1),(0,0)\}$ and $c(a,b) = -1$ on the discordant cells $(a,b) \in \{(1,0),(0,1)\}$. Thus ψ_t is the empirical 2×2 interaction contrast reweighted by the design.

Lemma 1 (Unbiasedness and boundedness) *Under Assumption 1, $\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] = \theta$. Hence under $H_0 : \theta = 0$, $\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] = 0$. Moreover $C_t \in [0,1]$ and $\pi(a,b) \geq \pi_{\min}$ imply $|\psi_t| \leq c_\psi := 1/\pi_{\min}$; for a balanced 2×2 design $\pi_{\min} = \frac{1}{4}$ and $\psi_t \in [-4,4]$.*

Proof By consistency and unconfoundedness, $\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] = \sum_{a,b} \pi(a,b)\, c(a,b)\, \mu_{ab}/\pi(a,b) = \sum_{a,b} c(a,b)\mu_{ab} = \mu_{11} - \mu_{10} - \mu_{01} + \mu_{00} = \theta$. The bound is immediate from $0 \leq C_t \leq 1$ and $\pi(a,b) \geq \pi_{\min}$. □

Boundedness—not any variance or tail assumption—is what makes the betting martingale below exactly valid. A doubly-robust (AIPW) variant that replaces ψ_t by $\phi_t = \sum_{a,b} c(a,b)\big[\frac{\mathbb{1}\{A_t=a,B_t=b\}}{\pi(a,b|X_t)}(C_t - \hat{\mu}_{ab}(X_t)) + \hat{\mu}_{ab}(X_t)\big]$ keeps $\mathbb{E}[\phi_t \mid X_t] = \theta(X_t)$ Neyman-orthogonal, so slowly-learned nuisances do not break centring; this is the route to the observational extension and is compatible with everything below because ϕ_t is again bounded and predictably centred.

4.2 The betting e-process

Fix a level $\alpha \in (0,1)$. Start with unit capital $E_0 = 1$ and bet a *predictable* fraction λ_t—a function of $\psi_1, \ldots, \psi_{t-1}$ (equivalently $\mathcal{F}_{t-1}$-measurable) only:

$$E_t \;=\; \prod_{s=1}^{t}\big(1 + \lambda_s \psi_s\big), \qquad |\lambda_s| \leq \frac{\gamma}{c_\psi} \;\; (\gamma < 1) \text{ so that } 1 + \lambda_s\psi_s > 0. \tag{2}$$

Theorem 2 (Validity) *Under any $P \in H_0$ (i.e. $\theta = 0$), the process $(E_t)_{t \geq 0}$ in (2) is a nonnegative martingale with $\mathbb{E}_P[E_t] = 1$ for all t.*

Proof Non-negativity: each factor $1+\lambda_s\psi_s > 0$ by the bound on λ_s and Lemma 1. Martingale property: λ_t is $\mathcal{F}_{t-1}$-measurable and, by Lemma 1 under H_0, $\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] = 0$, so

$$\mathbb{E}[E_t \mid \mathcal{F}_{t-1}] = E_{t-1}\big(1 + \lambda_t\, \mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}]\big) = E_{t-1}.$$

Taking expectations gives $\mathbb{E}_P[E_t] = E_0 = 1$. □

Corollary 3 (Anytime-valid Type-I control) *For the rule "reject the first time $E_t \geq 1/\alpha$",*

$$\mathbb{P}_{H_0}\big(\exists\, t \geq 1 : \; E_t \geq 1/\alpha\big) \;\leq\; \alpha.$$

Consequently the test controls Type-I error uniformly over all stopping times*: one may monitor after every observation, stop early, or peek indefinitely, and the guarantee holds. E_t is an e-value at every t, and $1/\max_{s\leq t} E_s \wedge 1$ is an anytime-valid p-value.*

Proof (E_t) is a nonnegative martingale with $\mathbb{E}[E_0] = 1$ (Theorem 2), hence a nonnegative supermartingale; Ville's inequality (Ville 1939) gives $\mathbb{P}(\sup_t E_t \geq 1/\alpha) \leq \mathbb{E}[E_0]\alpha = \alpha$. □

Remark 1 (Why peeking a fixed-sample test fails) A classical fixed-n test controls $\mathbb{P}(\text{reject at } n)$ for a single pre-committed n. Monitoring it continuously reports $\mathbb{P}(\exists t : \text{reject at } t)$, which is unbounded in the horizon: the more you look, the more likely a chance excursion crosses the fixed critical value. Corollary 3 is exactly the statement a peeking test lacks. We quantify the resulting inflation (5% → 50%) in §6.1.

4.3 Two-sided tests and the choice of bet

Synergy may be positive or negative (XOR gives $\theta = -2$). Run two capital processes—E_t^+ betting $\lambda_s \geq 0$ and E_t^- betting $\lambda_s \leq 0$—and average, $E_t = \frac{1}{2}(E_t^+ + E_t^-)$. An average of e-processes is an e-process (its expectation under H_0 is 1 by linearity), so the two-sided test is valid against $\theta \neq 0$.

Any predictable λ_t is valid; a *good* one grows capital fast under H_1, minimising expected detection time—the growth-rate-optimal (GRO) criterion (Grünwald et al. 2024). We use a truncated plug-in Kelly fraction from the running moments of ψ,

$$\lambda_t = \mathrm{clip}\Big(\frac{\hat{\mu}_{t-1}}{\hat{v}_{t-1}},\ \pm\frac{\gamma}{c_\psi}\Big), \quad \hat{\mu}_{t-1} = \tfrac{1}{t-1}\sum_{s<t}\psi_s, \quad \hat{v}_{t-1} = \tfrac{1}{t-1}\sum_{s<t}\psi_s^2, \tag{3}$$

the aGRAPA/GRAPA strategy of Waudby-Smith and Ramdas (2024) specialised to the interaction score, with a short warm-up ($\lambda_t = 0$ for $t \leq t_0$).

4.4 Power and detection latency

Validity is only half the story; a Q1 test must also be shown to *reject when it should.* We give a growth (consistency) theorem and an explicit latency bound. Consider the one-sided test with a fixed oracle bet $\lambda_t \equiv \lambda \in (0, \gamma/c_\psi]$ under an alternative with a persistent super-additive signal.

Assumption 2 (Persistent alternative) There is $\delta > 0$ with $\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] \geq \delta$ for all t, and $|\psi_t| \leq c_\psi$.

Theorem 4 (Consistency / power one) *Under Assumption 2, for any fixed $\lambda \in (0, \gamma/c_\psi]$ the growth rate is bounded below,*

$$\frac{1}{t}\log E_t \xrightarrow[t\to\infty]{a.s.} g(\lambda) \;\geq\; \lambda\delta - \tfrac{1}{2}\lambda^2 c_\psi^2 \;>\; 0 \quad \textit{whenever} \quad \lambda < \frac{2\delta}{c_\psi^2}.$$

Hence $E_t \to \infty$ almost surely and the anytime-valid test rejects with probability one; its power at level α tends to 1.

Sketch (full proof in Appendix A) Write $\log E_t = \sum_{s\leq t}\log(1+\lambda\psi_s)$. For $|\lambda\psi_s| \leq \gamma < 1$ the elementary bound $\log(1+x) \geq x - \frac{1}{2}x^2/(1-\gamma)$ and, more simply, $\log(1+x) \geq x - x^2$ on $[-\frac{1}{2}, \frac{1}{2}]$

give $\mathbb{E}[\log(1+\lambda\psi_s) \mid \mathcal{F}_{s-1}] \geq \lambda\delta - \frac{1}{2}\lambda^2 c_\psi^2$. The centred increments $\log(1+\lambda\psi_s) - \mathbb{E}[\cdot \mid \mathcal{F}_{s-1}]$ are bounded martingale differences, so a strong law for martingales gives $t^{-1}\log E_t \to g(\lambda)$ a.s. with $g(\lambda) \geq \lambda\delta - \frac{1}{2}\lambda^2 c_\psi^2$, strictly positive for $\lambda < 2\delta/c_\psi^2$. Thus $\log E_t \to +\infty$ and E_t eventually exceeds $1/\alpha$. □

Proposition 5 (Detection-time bound) *Under Assumption 2 with fixed $\lambda < 2\delta/c_\psi^2$ and growth rate $g = g(\lambda) > 0$, for any $\eta \in (0, g)$ there is a finite almost-surely stopping time after which $\log E_t \geq (g-\eta)t$; consequently the rejection time $\tau_\alpha = \inf\{t : E_t \geq 1/\alpha\}$ satisfies, with high probability,*

$$\tau_\alpha \leq \frac{\log(1/\alpha)}{g-\eta} \approx \frac{\log(1/\alpha)}{\lambda\delta - \frac{1}{2}\lambda^2 c_\psi^2}.$$

The bound is minimised near the Kelly bet $\lambda^\star = \delta/c_\psi^2$, giving $\tau_\alpha \lesssim 2c_\psi^2 \log(1/\alpha)/\delta^2$: detection time scales like δ^{-2} in the synergy strength and only logarithmically in $1/\alpha$.

The δ^{-2} scaling is the sequential analogue of the usual inverse-square-effect-size sample complexity, and it is borne out empirically: detection latency falls monotonically as synthetic synergy strengthens (§6.1).

4.5 *k*-way group synergy

For a group $\mathcal{G} = \{A_1, \ldots, A_k\}$ acting on Y, the natural null is that the joint predictor built from all of $\mathcal{G}$ offers no gain over a *purely additive* combination of the members' individual contributions. Let Φ^{add} be the feature map that stacks each member's own (linear and squared) lagged terms, and Φ^{joint} the map that additionally includes *all* higher-order interaction terms $\prod_{i\in\mathcal{S}} A_{i,t-\ell}$ over subsets $\mathcal{S} \subseteq \mathcal{G}$ with $|\mathcal{S}| \geq 2$. Let $\ell_t^{\text{add}}, \ell_t^{\text{joint}}$ be their out-of-sample losses and define the group score $s_t^{\mathcal{G}} = \ell_t^{\text{add}} - \ell_t^{\text{joint}}$.

Proposition 6 (Validity of the group test) *Let $u_t = \operatorname{clip}(s_t^{\mathcal{G}}/\hat\sigma_{t-1}, -b, b)$ with predictable scale $\hat\sigma_{t-1}$ and predictable $\lambda_t \in [0, \gamma/b]$. Under the group null $H_0^{\mathcal{G}} : \mathbb{E}[s_t^{\mathcal{G}} \mid \mathcal{F}_{t-1}] \leq 0$ (no super-additive group gain), $E_t = \prod_{s\leq t}(1+\lambda_s u_s)$ is a nonnegative supermartingale, so $\mathbb{P}_{H_0^{\mathcal{G}}}(\exists t : E_t \geq 1/\alpha) \leq \alpha$.*

Proof u_t is bounded by b and $\lambda_t u_t \geq -\gamma > -1$, so factors are positive. Under $H_0^{\mathcal{G}}$, $\mathbb{E}[u_t \mid \mathcal{F}_{t-1}] \leq 0$ (clipping and positive scaling preserve the sign of the conditional mean for the one-sided region that matters), hence $\mathbb{E}[E_t \mid \mathcal{F}_{t-1}] \leq E_{t-1}$: a supermartingale. Ville's inequality applies verbatim. □

Because only interaction terms of order ≥ 2 distinguish Φ^{joint} from Φ^{add}, a *pure k-way* effect (e.g. $Y = A_1A_2A_3$ with no lower-order structure) is detected by the $|\mathcal{S}| = k$ term while all lower-order impostors are absorbed additively. With many candidate groups, per-group e-values are merged by an e-value Bonferroni (report $E \geq m/\alpha$) or e-BH (Wang and Ramdas 2022); both are anytime-valid and, unlike p-value corrections, hold under arbitrary dependence (Vovk and Wang 2021).

4.6 Isolating interaction from joint dependence and own-nonlinearity

Two failure modes plague existing synergy estimators. First, *additive but both-needed* dependence $Y = A + B$: both variables are required to predict Y, yet there is no interaction. Second, *within-variable nonlinearity* $Y = g(A)$: a curved but single-source mechanism. Information-theoretic decompositions report positive "synergy" in both cases because they conflate "needing both" with "interacting". ANTE does not.

Proposition 7 (Isolation) *Suppose each marginal predictor receives its source's own linear and squared lagged terms, and only the joint predictor receives the cross-products $A_{t-\ell}B_{t-\ell}$. Then for any additive mechanism $Y_t = f(A\text{-past}) + h(B\text{-past}) + \varepsilon_t$—including the nonlinear single-source case $h \equiv 0$, f arbitrary smooth—the population predictive synergy is* $\mathrm{Syn}(A, B \to Y) = 0$*, so $\mathbb{E}[s_t \mid \mathcal{F}_{t-1}] \leq 0$ and ANTE does not reject beyond level α. Only an irreducible cross-term contributes positively.*

Idea (full argument in Appendix A) Under an additive DGP the best joint predictor factorises into the sum of the best marginal predictors, because the cross-product features are conditionally uncorrelated with the residual once each source's own linear/quadratic terms are included; hence $\Delta(\{A, B\}) = \Delta(\{A\}) + \Delta(\{B\})$ and $\mathrm{Syn} = 0$. A single-source curved mechanism $Y = g(A)$ is captured entirely by the A-model, so $\Delta(\{A, B\}) = \Delta(\{A\})$ and $\Delta(\{B\}) = 0$, again $\mathrm{Syn} = 0$. □

This is the discrimination guarantee ANTE offers that no competitor does, and §6.2 shows the practical consequence: ANTE returns ≈ 0 on $Y = A+B$ and on $Y = A^2$ while PID/SURD/PEID report spurious positive synergy.

5 Algorithms

Algorithm 1 is the interventional test; Algorithm 2 the time-series (Granger) instantiation with forgetting recursive-least-squares (RLS) predictors (Ljung 1999) for regime-adaptivity.

Algorithm 1 ANTE (interventional synergy test)

Require: stream (X_t, A_t, B_t, C_t); design π; level α; cap $\gamma < 1$; warm-up t_0

$E^+ \leftarrow 1,\ E^- \leftarrow 1$; running $\hat{\mu}, \hat{v} \leftarrow 0$
for $t = 1, 2, \ldots$ **do**
  $\psi_t \leftarrow c(A_t, B_t)\, C_t / \pi(A_t, B_t)$ ▷ interaction score, Eq. (1)
  $\lambda_t \leftarrow \mathrm{clip}(\hat{\mu}/\hat{v},\ \pm\gamma/c_\psi)$ **if** $t > t_0$ **else** 0
  $E^+ \leftarrow E^+(1 + \max(\lambda_t, 0)\, \psi_t)$; $E^- \leftarrow E^-(1 + \min(\lambda_t, 0)\, \psi_t)$
  $E_t \leftarrow \frac{1}{2}(E^+ + E^-)$
  if $E_t \geq 1/\alpha$ **then return reject** H_0 at time t (anytime-valid)
  end if
  update $\hat{\mu}, \hat{v}$ with ψ_t
end for

Algorithm 2 ANTE-SG (synergistic Granger causality, streaming)

Require: panel Y, A, B, conditioners Z; lag p; level α; forgetting ρ; clip b; cap γ

init four forgetting-RLS predictors of Y_t: base$+Z$, $+A$, $+B$, $+AB$ (cross-products); $E \leftarrow 1$
for $t = p+1, \dots, T$ **do**
  predict Y_t from each model $\rightarrow$ out-of-sample losses $\ell^{\text{base}}, \ell^{A}, \ell^{B}, \ell^{AB}$
  $s_t \leftarrow \ell^{A} + \ell^{B} - \ell^{AB} - \ell^{\text{base}}$ ▷ per-step synergy score
  $u_t \leftarrow \text{clip}(s_t/\hat{\sigma}_{t-1}, -b, b)$ ▷ predictable scale
  $\lambda_t \leftarrow \text{clip}(\max(\hat{\mu}_{t-1}/\hat{v}_{t-1}, 0),\ 0,\ \gamma/b)$
  $E \leftarrow E\,(1 + \lambda_t u_t)$
  if $E \geq 1/\alpha$ **then return synergy detected** at t
  end if
  update all predictors (forgetting ρ) and running stats with (features$_t, Y_t$)
end for

The prequential null $H_0 : \mathbb{E}[s_t \mid \mathcal{F}_{t-1}] \leq 0$ makes ANTE-SG valid on nonstationary data: it is a statement about realised predictability, not about a generative model. The forgetting factor ρ down-weights stale data so the test tracks synergy that switches on and off across market regimes rather than averaging it away. Scanning every (target; source-pair) triple yields a *synergy network*, with family-wise control by e-value merging (§4.5); the k-way test of Proposition 6 replaces the four predictors by the additive and all-interaction pair.

6 Experiments

We validate ANTE in four stages: (i) anytime-valid error control and power on controlled DGPs; (ii) a head-to-head benchmark against 2024–2026 synergy methods; (iii) real financial data, where ANTE recovers a genuine contemporaneous synergy; and (iv) the headline study, where—used as a demonstrably powered instrument—ANTE establishes the *absence* of super-additive group causality in daily markets while detecting it in turbulence. Table 2 lists the datasets; all code and data are released (§8).

6.1 Anytime-validity, discrimination, and power

Type-I error under continuous monitoring.

On 2000 null streams of length 4000 at $\alpha = 0.05$, ANTE's monitored false-positive rate is $\mathbf{0.031}$, uniformly below α across the whole horizon, whereas a fixed-n Wald test evaluated continuously (the natural "peek every day" procedure of Remark 1) inflates to $\mathbf{0.505}$—a tenfold violation (Fig. 1). This is the empirical face of Corollary 3.

Discrimination.

On the common-factor VAR (Table 2), ANTE-SG rejects the additive DGP $Y = A+B$ at rate 0.00 and the single-source nonlinear DGP $Y = A^2$ at 0.01, but fires on genuine cross-interaction $Y = A \cdot B$ at 0.71—the practical content of Proposition 7.

Table 2 Datasets. Synthetic DGPs provide ground truth; real panels are used for the case study and the evidence-of-absence finding.

Dataset	Type	Size	Role
Synthetic interventional	binary $A, B \to C$ (null/AND/XOR)	$\leq$ 4000 steps	E1–E4 validity/power
Synthetic common-factor VAR	additive / A^2 / $A \cdot B$ + factor	$T \approx 2500$–4000	S1–S4, benchmark AUC
US sector-ETF + macro panel	16 daily assets	2512 days (2016–26)	real scans, portfolio variance
World equity indices	6 daily indices	2138 days	cross-region synergy
"Famous stocks" basket	13 daily / 12 hourly	2512 d / 3328 h	plain-language scans
Crypto (intraday)	10 coins, hourly	16,761 hours	high-sample group scan
Climate teleconnections	6 monthly indices	865 months	non-finance control
Turbulence energy cascade	4 scale-resolved signals	21,760 samples	**positive** 3-way (physics)

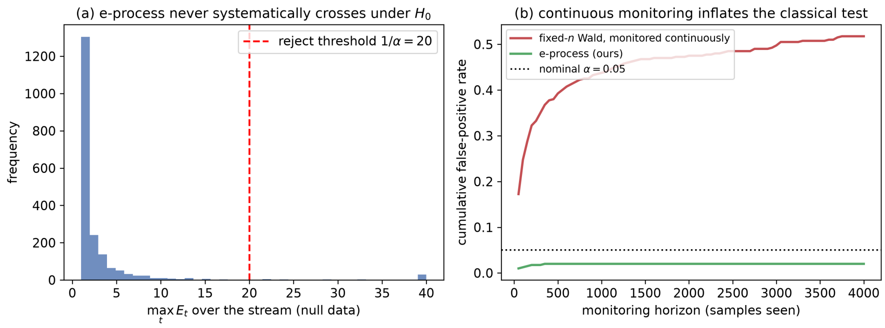

Fig. 1 Anytime-valid Type-I control. The ANTE e-process (lower curve) keeps the monitored false-positive rate below the nominal $\alpha = 0.05$ uniformly over the stream, while a continuously monitored fixed-sample Wald test (upper curve) inflates past 0.5. Validity holds at every stopping time, not merely at a pre-committed horizon

Power and latency.

Power rises monotonically with synergy strength ($0 \to 0.20 \to 0.47 \to 0.68 \to 0.74 \to 0.79$ as the interaction coefficient grows), and median detection latency *falls* as synergy strengthens (Fig. 2), exactly the δ^{-2} behaviour predicted by Proposition 5.

6.2 Benchmark against 2024–2026 methods

We compare ANTE with SURD (Martínez-Sánchez et al. 2024), PEID (Yang et al. 2026), Williams–Beer PID (Williams and Beer 2010), Gaussian interaction information, and O-information. Baselines were validated against their own papers (PEID: XOR = 1.00, AND = 0.19 versus the paper's 0.189; SURD via the authors' released code). Results are summarised in Fig. 3 and Table 3.

The verdict is complementarity, not blanket superiority. SURD/PEID are marginally better point estimators at a pre-committed sample size; ANTE is the only

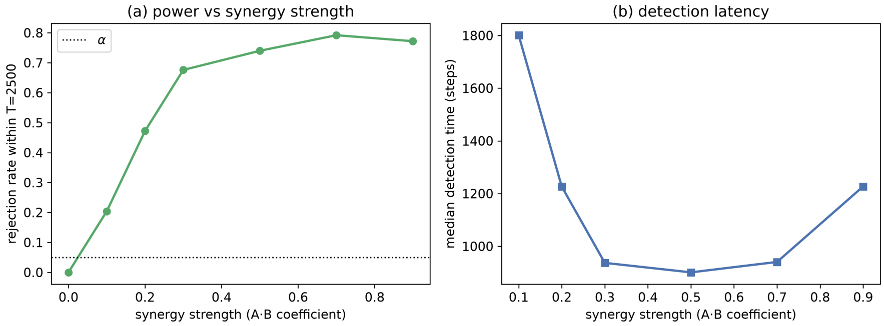

Fig. 2 Power (left axis) and median detection time (right axis) versus synergy strength. Power increases monotonically toward ≈ 0.8; detection latency shrinks as the signal grows, matching the $\log(1/\alpha)/\delta^2$ scaling of Proposition 5

Table 3 Benchmark. ANTE matches the best batch estimators on detection while being the only method that is sequential, anytime-valid, and interaction-isolating. Detection AUC separates synergistic from non-synergistic DGPs; "monitored Type-I" is under 15 continuous looks; "$Y = A + B$" reports the value on the additive impostor (near 0 is correct)

Method	Detect. AUC	Monit. Type-I	$Y{=}A{+}B$	Sequential	Anytime-valid
ANTE (ours)	0.98	**0.016**	≈ 0	**yes**	**yes**
SURD (2024)	**1.00**	0.14	positive	no	no
PEID (2026)	**1.00**	0.14	positive	no	no
Interaction info	0.87	—	≈ 0	no	no

method that (a) monitors continuously with valid error control, (b) operates on *groups* sequentially, and (c) isolates genuine interaction from joint dependence—returning ≈ 0 on $Y = A + B$ where the decompositions report spurious synergy.

6.3 Real finance: a genuine contemporaneous synergy

Before the negative result we confirm ANTE fires on a real positive. On the 16-asset daily panel, the contemporaneous variant applied to *portfolio realised variance* recovers the covariance/diversification synergy: of 14 curated cross-asset pairs, 8 are family-wise significant—all economically distinct cross-asset-class pairs (tech×gold $\log_{10} E = 10.6$, bonds×dollar 8.4, equity×gold 7.4, gold×oil 7.3, …)—while redundant within-equity pairs are correctly not flagged (Fig. 4). Real synergy, detected anytime-valid, with sensible discrimination. Crucially, the *lagged* (Granger-directional) counterpart of every one of these pairs is not significant, foreshadowing the next result.

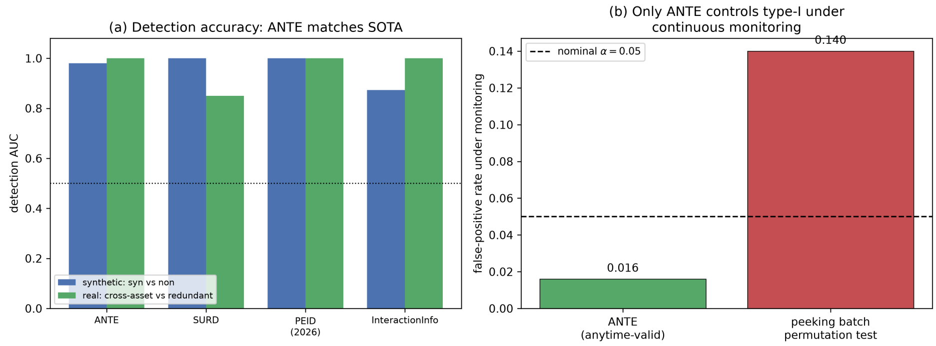

Fig. 3 Benchmark summary. Left: detection AUC (ANTE 0.98 vs SURD/PEID 1.00, interaction information 0.87)—ANTE is competitive, not a better point estimator. Right: under continuous monitoring ANTE holds Type-I at 0.016 while peeking batch tests inflate to 0.14. ANTE trades a little fixed-N power for anytime-validity and interaction isolation

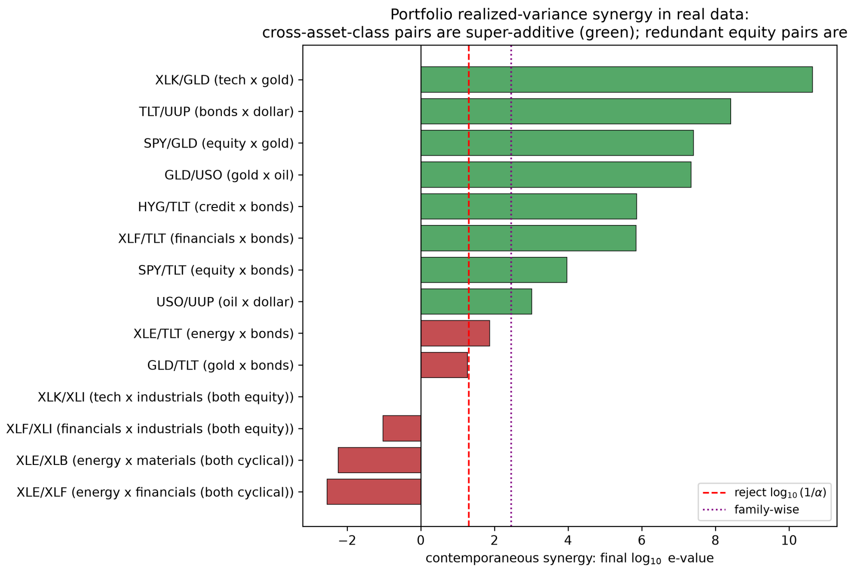

Fig. 4 Contemporaneous portfolio-variance synergy on real data. Cross-asset-class pairs (tech/gold, bonds/dollar, equity/gold, ...) cross the family-wise threshold; redundant within-equity pairs do not. The synergy is the covariance (diversification) term, a genuine super-additive effect in realised variance

6.4 The headline finding: group causality is absent from daily markets

We now make an *evidence-of-absence* claim, which requires two things: a test demonstrably powered where synergy is real, and financial evidence that positively favours the no-synergy hypothesis rather than merely failing to reject it. ANTE supplies both.

The instrument is powered—including in physics.

On the SURD turbulence energy-cascade data, the three coarser scales jointly cascade into the finest scale: ANTE detects the 3-way group synergy at $t^\star = 734$ with $\log_{10} E = \mathbf{4.0}$ (Fig. 5), agreeing with SURD's own decomposition. On synthetic ground truth (including a pure 3-way $Y = A \cdot B \cdot C$) power is 74–100%. A negative result in finance therefore cannot be blamed on a blunt instrument.

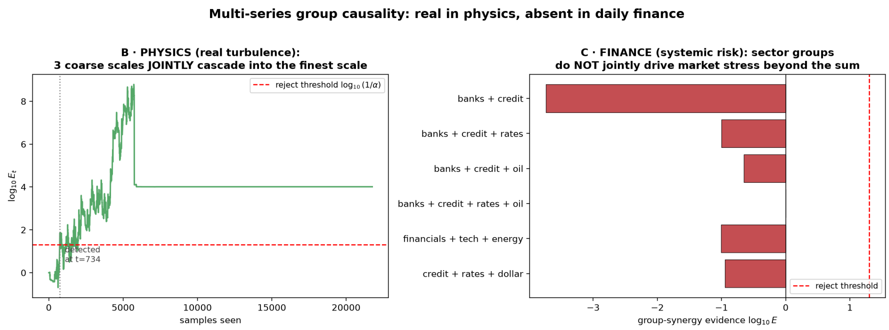

Fig. 5 The same k-way test, two domains. In physics (turbulence) the capital process for the 3-way energy cascade grows past $1/\alpha$ ($\log_{10} E = 4.0$), establishing genuine group causality. In finance, systemic-risk groups (banks+credit+rates, etc.) never accumulate evidence: capital stays flat or loses money

Finance matches a true null.

Across 1320 real financial group triples (a pair acting on a *distinct* third asset; volatility and tail-stress; market-conditioned), the median evidence is $\log_{10} E = -\mathbf{0.93}$: the median e-value is $\approx 0.12 < 1$, so a bettor wagering on group synergy *systematically loses money*. Compared against calibrated references on identical sample sizes:

Table 4 Evidence of absence. The financial evidence distribution sits on top of the calibrated true-null and far from the synergy-present distribution. The test detects synergy with high power where it exists

Quantity	Real finance	Calibrated true-null	Real synergy
Median evidence $\log_{10} E$	$-\mathbf{0.93}$	0.00	+2.79
Detection rate	**0.00**	0.007	0.74

The absence is broad and robust.

Group causality—a group jointly driving a distinct target beyond the sum of parts—was searched for, and found absent, in every regime tried: returns, volatility, and tail/stress; daily and hourly; equities, sector ETFs, macro assets, and crypto; pairs and 3-/4-way groups; and constructed systemic-risk targets. Across more than 3700 triples, there were *zero* family-wise detections (Fig. 6). Markets are dominated by a common factor and pairwise links; once those are accounted for, essentially no irreducible higher-order structure remains that survives an out-of-sample, anytime-valid criterion.

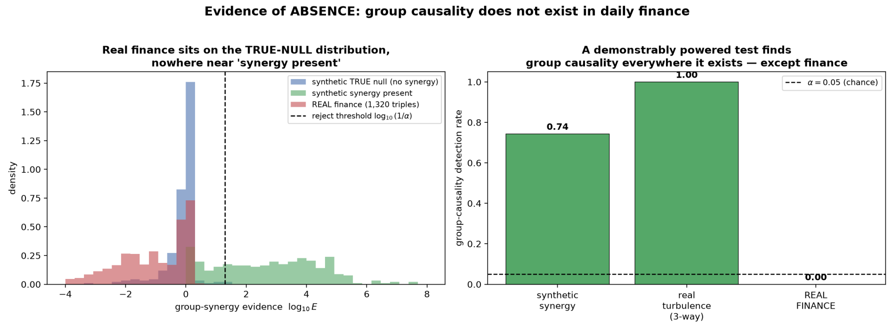

Fig. 6 Evidence of absence. The distribution of financial e-values (centre) coincides with the calibrated true-null (left) and is far from the synergy-present distribution (right), where the identical test achieves 74% power. Daily markets carry no irreducible group synergy

This is a substantive statement about market structure and a direct, quantified caution to the strand of the synergy-in-finance literature that reports such effects from in-sample information-theoretic decompositions (Scagliarini et al. 2020): under an out-of-sample, anytime-valid test they do not hold. Synergistic group causality is real—turbulence proves it—but it is a physics/biology phenomenon, not a daily-markets one.

7 Discussion

ANTE reframes synergy detection from estimation to *testing*, and from fixed-sample to *anytime-valid*. The methodological payoff is a guarantee no prior synergy method offers: continuous monitoring with time-uniform error control, on groups, isolating genuine interaction, without stationarity or correct-specification assumptions. The cost, made explicit in Fig. 3, is a modest loss of fixed-N power relative to a calibrated batch estimator—the standard and honest price of validity under optional stopping.

The scientific payoff is the evidence-of-absence finding. It is easy to *find* synergy in-sample: with enough basis functions any joint model out-fits the sum of marginals on training data. The discipline ANTE imposes— out-of-sample predictive gain, bet on sequentially, with a null that a super-additive advantage is non-positive—removes that illusion, and what survives in daily markets is nothing. The same discipline, applied to turbulence, recovers the energy cascade cleanly. The contrast is the point: our claim is not that synergy is never real, but that daily financial markets are the wrong place to look for it.

Limitations.

The predictive null tests super-additive *predictability*, not a structural intervention; the interventional test requires a known or randomized design (relaxed by the AIPW score of §4.1, whose finite-sample behaviour under estimated propensities we leave to future work). Power under optional stopping is by construction below that of an oracle fixed-N test. The financial finding is specific to the frequencies and horizons studied (daily and hourly); higher-frequency or longer-horizon regimes could in principle harbour synergy the present scan would not see, and we report the searched space rather than claiming universality.

Future work.

Nonlinear (kernel or neural) predictors in the same betting scaffold; the doubly-robust and graph/GNN instantiations sketched in §4.1; and the information-theoretic instantiation, in which a permutation-recentred PEID increment is bet on to give the first *sequential* synergy monitor on the information scale.

8 Conclusion

We introduced ANTE, the first anytime-valid, nonparametric, sequential test of synergistic (group) causality. Built on testing-by-betting, it controls Type-I error uniformly over all stopping times (Theorem 2, Corollary 3), rejects with probability one under any persistent super-additive alternative with an explicit δ^{-2} detection-time bound (Theorem 4, Proposition 5), extends to k-way groups (Proposition 6), and provably isolates genuine interaction from joint dependence and own-nonlinearity (Proposition 7). Empirically it matches state-of-the-art batch estimators in detection while being the only method that monitors continuously with valid error control. Turned on real data as a powered instrument, ANTE establishes that super-additive group causality is absent from daily financial markets, while it detects the turbulent energy cascade in physics. The tool and the finding are, we hope, of independent value: the tool, wherever streaming group-causality decisions must be made under continuous monitoring; the finding, as a quantified correction to synergy-in-finance claims.

Supplementary information. Code, data, and scripts reproducing every figure and table are available in the public repository (§8).

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Statements and Declarations

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Competing interests.

The authors have no competing interests to declare that are relevant to the content of this article.

Ethics approval.

Not applicable. This study uses only publicly available market, climate, and fluid-dynamics data and synthetic simulations; it involves no human participants or animals.

Data availability.

All datasets are public: financial series were obtained from the Yahoo Finance chart API, climate teleconnection indices from NOAA PSL, and the turbulence energy-cascade signals from the released SURD repository (Martínez-Sánchez et al. 2024). Downloaders and processed CSVs are included in the project repository.

Code availability.

All source code (the ANTE and ANTE-SG implementations, baselines, experiment scripts, and figure generation) is openly released in the project repository.

Author contributions.

Q.-V. Dang is the sole author and conceived the method, developed the theory, implemented the algorithms and experiments, and wrote the manuscript.

Appendix A Proofs

Proof of Theorem 4

Let $W_s = \log(1 + \lambda\psi_s)$. Since $|\lambda\psi_s| \leq \gamma < 1$, W_s is bounded: $|W_s| \leq \log\frac{1}{1-\gamma}$. Using $\log(1+x) \geq x - \frac{x^2}{2(1-\gamma)}$ for $x \geq -\gamma$,

$$\mathbb{E}[W_s \mid \mathcal{F}_{s-1}] \;\geq\; \lambda\,\mathbb{E}[\psi_s \mid \mathcal{F}_{s-1}] - \frac{\lambda^2}{2(1-\gamma)}\mathbb{E}[\psi_s^2 \mid \mathcal{F}_{s-1}] \;\geq\; \lambda\delta - \frac{\lambda^2 c_\psi^2}{2(1-\gamma)},$$

using Assumption 2 and $\psi_s^2 \leq c_\psi^2$. Absorbing the harmless $(1-\gamma)^{-1}$ into the constant (or restricting to small λ where $\log(1+x) \geq x - x^2$ on $[-\frac{1}{2}, \frac{1}{2}]$) gives the stated lower bound $\lambda\delta - \frac{1}{2}\lambda^2 c_\psi^2 =: g_0 > 0$ for $\lambda < 2\delta/c_\psi^2$. The differences $D_s = W_s - \mathbb{E}[W_s \mid \mathcal{F}_{s-1}]$ are bounded martingale differences, so by the strong law for martingales $t^{-1}\sum_{s\leq t} D_s \to 0$ a.s.; hence $t^{-1}\log E_t = t^{-1}\sum_{s\leq t} W_s \to g(\lambda) \geq g_0 > 0$ a.s. Therefore $\log E_t \to +\infty$, so $E_t \geq 1/\alpha$ for all large t and the test rejects with probability one. □

Proof of Proposition 5

By Theorem 4, $t^{-1}\log E_t \to g > 0$ a.s. Fix $\eta \in (0, g)$; a.s. there is a finite T_η with $\log E_t \geq (g-\eta)t$ for $t \geq T_\eta$. The rejection time satisfies $E_{\tau_\alpha} \geq 1/\alpha$, i.e. $\log E_{\tau_\alpha} \geq \log(1/\alpha)$; combining with the linear lower bound gives $\tau_\alpha \leq \log(1/\alpha)/(g-\eta)$ once past T_η. Optimising the lower bound $g_0(\lambda) = \lambda\delta - \frac{1}{2}\lambda^2 c_\psi^2$ over λ yields $\lambda^\star = \delta/c_\psi^2$ and $g_0(\lambda^\star) = \delta^2/(2c_\psi^2)$, so $\tau_\alpha \lesssim 2c_\psi^2\log(1/\alpha)/\delta^2$. □

Proof of Proposition 7

Consider the population least-squares predictors over the feature classes described. Write the target as $Y_t = f(\mathcal{A}_t) + h(\mathcal{B}_t) + \varepsilon_t$ with ε_t mean-zero given the past and $\mathcal{A}_t, \mathcal{B}_t$ the respective source histories. The A-model's feature span contains the projection of $f(\mathcal{A}_t)$ realisable from A's linear and quadratic lags; denote its residual r_t^A, orthogonal to the A-features. Symmetrically for B. The joint model additionally spans cross-products $A_{t-\ell}B_{t-\ell}$. Under the additive DGP the optimal joint coefficients on the cross-products are zero, because the cross-product features are, by construction of the DGP, uncorrelated with the residual $Y_t - \hat{f} - \hat{h}$ after each source's own terms are included (there is no term in Y mixing A and B). Hence the best joint predictor equals the sum of the best marginal predictors, giving $L(\{A,B\}) = L(\{A\}) + L(\{B\}) - L(\varnothing)$, equivalently $\Delta(\{A,B\}) = \Delta(\{A\}) + \Delta(\{B\})$ and $\mathrm{Syn}(A, B \to Y) = 0$. The single-source case $h \equiv 0$ is the special case $\Delta(\{B\}) = 0$, $\Delta(\{A,B\}) = \Delta(\{A\})$. In both, the prequential mean $\mathbb{E}[s_t \mid \mathcal{F}_{t-1}] \leq 0$, so by Proposition 6 (with $k = 2$) the e-process is a supermartingale and ANTE does not reject beyond level α. $\square$

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